

$$\textcircled{10} \langle n | g(t) | m \rangle = \frac{1}{2} (N_h + 1) (2 \langle n | a^\dagger g a^\dagger | m \rangle - \langle n | a^\dagger a g | m \rangle - \langle n | g a^\dagger a | m \rangle) \\ + \frac{1}{2} N_h (2 \langle n | a^\dagger g a | m \rangle - \langle n | a a^\dagger g | m \rangle - \langle n | g a a^\dagger | m \rangle)$$

$$g(t) = \exp(-i/\hbar t H) | \psi \rangle \langle \psi | \exp(i/\hbar t H^\dagger) \quad a a^\dagger - a^\dagger a = 1 \quad a a^\dagger = 1 + a^\dagger a \\ \langle n | a g a^\dagger | m \rangle (2\delta_{N_h+1} + \delta) - (\delta_{N_h+1} + \frac{\delta}{2}) [\langle n | a^\dagger a g | m \rangle - \langle n | g a^\dagger a | m \rangle] \\ = \langle n | a g a^\dagger | m \rangle (2\delta_{N_h+1} + \delta) - \delta (N_h + \frac{1}{2}) [\langle n | a^\dagger a g | m \rangle - m \langle n | g | m \rangle]$$

$$\textcircled{1} \frac{1}{2} \langle n | a (|n\rangle + |m\rangle) (\langle n| + \langle m|)^\dagger | m \rangle = \langle n | (\frac{1}{2} |n\rangle + \frac{1}{2} |m\rangle) | m \rangle = \langle n | (\frac{1}{2} |n\rangle + \frac{1}{2} |m\rangle) | m \rangle$$

$$\textcircled{2} \langle n | a^\dagger a \frac{1}{2} (|n\rangle + |m\rangle) (\langle n| + \langle m|) | m \rangle = \frac{1}{2} (n + m)$$

$$\textcircled{3} \frac{1}{2} \langle n | (|n\rangle + |m\rangle) (\langle n| + \langle m|) | m \rangle = \frac{1}{2}$$

$$\textcircled{4} \textcircled{2} + \textcircled{3} = -\frac{1}{2} (N_h + \frac{1}{2}) \delta_{n,m} = -\frac{1}{2} n (N_h + \frac{1}{2})$$

$$\exp[-i/\hbar t \hbar \omega (a^\dagger a + \frac{1}{2})] = \sum_{n=0}^{\infty} \frac{[-i t \omega (a^\dagger a + \frac{1}{2})]^n}{n!}$$

$$\text{since } [C, \frac{1}{2}, [C, \frac{1}{2}, C a^\dagger a]] = 0 \text{ we have } \exp[-i/\hbar t \hbar \omega (a^\dagger a + \frac{1}{2})] = \exp[C/2] \exp[C a^\dagger a]$$

$$\hookrightarrow \exp[C (a^\dagger a + \frac{1}{2})] = \exp(C/2) \sum_{n=0}^{\infty} \frac{[-i t \omega a^\dagger a]^n}{n!}$$

$$g = \left[\exp(C/2) \sum_{n=0}^{\infty} \frac{[-i t \omega a^\dagger a]^n}{n!} \right] \cdot \frac{1}{2} (|n\rangle + |m\rangle) (\langle n| + \langle m|) \left[\exp(C/2) \sum_{n=0}^{\infty} \frac{[-i t \omega a^\dagger a]^n}{n!} \right]^\dagger$$

$$= \frac{1}{2} \left[1 + \sum_{n=1}^{\infty} \frac{[-i t \omega]^n}{n!} [n^n + m^n] \right] (|n\rangle + |m\rangle) (\langle n| + \langle m|) \left[1 + \sum_{n=1}^{\infty} \frac{[-i t \omega]^n}{n!} [n^n + m^n] \right]$$

$$= \frac{1}{2} \left[1 + \sum_{n=1}^{\infty} \frac{[-i t \omega]^n}{n!} [n^n |n\rangle + m^n |m\rangle] \right] [\langle n| n^n + \langle m| m^n] \left[1 + \sum_{n=1}^{\infty} \frac{[-i t \omega]^n}{n!} [n^n + m^n] \right]$$

$$= \frac{1}{2} [n \times n + m \times m + e^{-i t \omega n} |m \times n| e^{i t \omega n} + e^{-i t \omega m} |n \times m| e^{i t \omega m}]$$

$$= \frac{1}{2} \begin{pmatrix} e^{-i t \omega (n-m)} & 1 \\ e^{-i t \omega (m-n)} & 1 \end{pmatrix}$$

$$\textcircled{1} \langle n | a^\dagger a | m \rangle = \frac{1}{2} \langle n | [\sqrt{n+1} | n-1 \rangle X_n + \sqrt{m} | m-1 \rangle X_m + \sqrt{m} e^{-it\omega_m} | m-1 \rangle X_n e^{it\omega_n} + \sqrt{n+1} e^{-it\omega_n} | n-1 \rangle X_m e^{it\omega_m}] | m \rangle = 0$$

$$\textcircled{2} \langle n | a^\dagger a | m \rangle = \frac{1}{2} \langle n | a^\dagger a (| m \rangle + | n \rangle) e^{-it\omega_n} e^{it\omega_m} = \frac{1}{2} n e^{it(\omega_m - \omega_n)}$$

$$\textcircled{3} \langle n | g a^\dagger a | m \rangle = \frac{1}{2} m \langle n | (| m \rangle + | n \rangle) e^{-it\omega_n} e^{it\omega_m} = \frac{1}{2} m e^{it(\omega_m - \omega_n)}$$

$$\hookrightarrow \dot{c}_{nm}(t) = -\gamma(2N_{th}+1) c_{nm}(t) - \gamma(N_{th}+1/2) \cdot \frac{1}{2} e^{it(\omega_m - \omega_n)} (n-n)$$

$$c_{nm}(t) = \langle n | g(t) | m \rangle = \frac{1}{2} [e^{-it\omega_m} e^{it\omega_n}] = \frac{1}{2} e^{it(\omega_m - \omega_n)}$$

$$\text{for } t \rightarrow 0: c_{nm}(0) = -\gamma(N_{th}+1/2) \cdot \frac{n-m}{2} \quad c_{nm}(0) = 1/2$$

$$\Rightarrow -\gamma(N_{th}+1/2) \frac{(n-m)}{2} \cdot c_{nm}(0)$$

$$\textcircled{4} \langle n | a^\dagger a | m \rangle = \langle n | (1+a^\dagger a) | m \rangle = \langle n | g | m \rangle + \frac{1}{2} n e^{it(\omega_m - \omega_n)} = \frac{1}{2} e^{it(\omega_m - \omega_n)} (1+n)$$

$$\textcircled{5} \langle n | g a^\dagger a | m \rangle = \frac{1}{2} e^{it(\omega_m - \omega_n)} (m+1)$$

$$\hookrightarrow \langle n | \dot{g} | m \rangle = -\left[\frac{1}{2} n + \frac{1}{2} m \right] \frac{1}{2} (N_{th}+1) - \frac{1}{2} N_{th} \left[\frac{1}{2} (1+n) + \frac{1}{2} (1+m) \right]$$

$$= -\frac{1}{2} N_{th} \left[-\frac{1}{2} n - \frac{1}{2} m - \frac{1}{2} - \frac{1}{2} n - \frac{1}{2} - \frac{1}{2} m \right] - \frac{1}{2} \left[\frac{1}{2} n + \frac{1}{2} m \right]$$

$$= -\frac{1}{2} \left(N_{th} (n+m+1) \gamma + \gamma \left[\frac{1}{2} n + \frac{1}{2} m \right] \right)$$

$$\langle n | g(0) | m \rangle = 1/2$$

$$\dot{c}_{nm}(t) = -\frac{\Gamma_{nm}}{2} c_{nm}(t) = -\frac{1}{2} \cdot \frac{1}{2} \cdot \gamma [2N_{th}(m+n+1) + (n+m)]$$

$$= -\Gamma_{nm}$$

$$\begin{aligned}
 \langle \dot{a}(t) \rangle &= \text{Tr} \langle a(t) \dot{g}(t) \rangle = \frac{i}{\hbar} \langle [H, a(t)] \rangle + \sum_k \frac{\Omega_k}{2} \langle [c_k^\dagger, a(t)] c_k + c_k^\dagger [a(t), c_k] \rangle \\
 \frac{i}{\hbar} \langle [H, a(t)] \rangle &= \frac{i}{\hbar} \langle -i\hbar \frac{g}{2} (a^\dagger a^2 - a a^{\dagger 2}) + i\hbar \Omega (-a^\dagger a + a a^\dagger) \rangle = \frac{g}{2} \langle a^\dagger a + a a^\dagger \rangle \\
 &= \frac{g}{2} \langle a^\dagger [a, a] + [a^\dagger, a] a^\dagger + i\hbar \Omega \rangle = -\frac{g}{2} \langle a^\dagger a + a a^\dagger \rangle = -\frac{g}{2} \langle a^\dagger + a \rangle \\
 \sum_k \frac{\Omega_k}{2} \langle [c_k^\dagger, a(t)] c_k + c_k^\dagger [a(t), c_k] \rangle &= \frac{\Omega}{2} \langle [a^\dagger, a] a + a^\dagger [a, a] \rangle + \frac{\Omega}{2} \langle a^\dagger + a \rangle \\
 \langle [a, a] a^\dagger + a [a, a^\dagger] \rangle &= \frac{\Omega}{2} \langle -a \rangle + \frac{\Omega}{2} \langle a \rangle = -\frac{\Omega}{2} \langle a \rangle \\
 \hookrightarrow \langle \dot{a}(t) \rangle &= -\frac{g}{2} \langle a^\dagger + a \rangle - \frac{\Omega}{2} \langle a \rangle
 \end{aligned}$$

$$\begin{aligned}
 \langle \dot{a}^\dagger(t) \rangle &= \frac{i}{\hbar} \langle [H, a^\dagger(t)] \rangle + \sum_k \frac{\Omega_k}{2} \langle [c_k^\dagger, a^\dagger(t)] c_k + c_k^\dagger [a^\dagger(t), c_k] \rangle \\
 \frac{i}{\hbar} \langle [H, a^\dagger(t)] \rangle &= \frac{i}{\hbar} \langle i\hbar \frac{g}{2} (a^2 a^\dagger - a^\dagger a^{\dagger 2}) + i\hbar \Omega [a, a^\dagger] \rangle \\
 &= \frac{g}{2} \langle a [a, a^\dagger] + a^\dagger [a, a^\dagger] a \rangle + \frac{\Omega}{2} \langle a^\dagger - a \rangle \\
 &= -\frac{g}{2} \langle a^\dagger + a \rangle + \frac{\Omega}{2} \langle a^\dagger - a \rangle
 \end{aligned}$$

$$\frac{\Omega}{2} (N+1) \langle a^\dagger [a, a] \rangle + \frac{\Omega}{2} N \langle [a, a^\dagger] a^\dagger \rangle = -\frac{\Omega}{2} (N+1) \langle a^\dagger \rangle + \frac{\Omega}{2} N \langle a^\dagger \rangle = -\frac{\Omega}{2} \langle a^\dagger \rangle$$

$$\hookrightarrow \langle a^\dagger \rangle = -\frac{g}{2} \langle a^\dagger + a \rangle - \frac{\Omega}{2} \langle a^\dagger \rangle$$

$$\frac{d}{dt} \begin{pmatrix} \langle a \rangle \\ \langle a^\dagger \rangle \end{pmatrix} = \begin{pmatrix} -g/2 & -g \\ -g & -g/2 \end{pmatrix} \begin{pmatrix} \langle a \rangle \\ \langle a^\dagger \rangle \end{pmatrix} + \frac{\Omega}{2} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$\frac{d}{dt} \begin{pmatrix} \langle a(t) \rangle \\ \langle a^\dagger(t) \rangle \end{pmatrix} = \begin{pmatrix} -g/2 & -g \\ g & -g/2 \end{pmatrix} \begin{pmatrix} \langle a(t) \rangle \\ \langle a^\dagger(t) \rangle \end{pmatrix} + \frac{\Omega}{2} \begin{pmatrix} -1 \\ 1 \end{pmatrix} \langle a(0) \rangle$$

$$b) g = -\Omega = 0$$

$$G^{(2)}(\tau) = \langle a^\dagger(0) a^\dagger(\tau) a(\tau) a(0) \rangle$$

$$G^{(2)}(0) = \langle a^{\dagger 2}(0) a^2(0) \rangle = \text{tr} \{ a^{\dagger 2} a^2 \rho \} = 2 N_{th}^2 = 2 \left(\frac{1}{e^{\frac{\hbar\omega}{kT}} - 1} \right)^2$$

$$G^{(2)}(\tau) = \text{tr} (a^\dagger(0) a^\dagger(\tau) a(\tau) a(0) \rho) = \text{tr} (a^\dagger(0) [a a^\dagger(\tau) - 1] a(0) \rho)$$

$$= -\text{tr} (a^\dagger(0) a(0) \rho) + \text{tr} (a^\dagger(0) a(\tau) a^\dagger(\tau) a(0) \rho)$$

$$\Rightarrow \langle a^\dagger a \rangle^2$$

$$\text{Isserlis' Theorem: } \langle X_1 \dots X_n \rangle = \sum_{\text{pairings}} \langle X_{i_1} X_{j_1} \rangle \dots \langle X_{i_k} X_{j_k} \rangle = \langle X_1 X_2 \rangle \langle X_3 X_4 \rangle + \langle X_1 X_3 \rangle \langle X_2 X_4 \rangle + \langle X_1 X_4 \rangle \langle X_2 X_3 \rangle$$

$$\text{For } X \in \mathbb{R}^d \text{ being a Gaussian vector}$$

with zero mean [given in the thermal equilibrium]

$$\hookrightarrow \langle a^\dagger(0) a^\dagger(\tau) a(\tau) a(0) \rangle = \langle a^\dagger(0) a(\tau) \rangle \langle a^\dagger(\tau) a(0) \rangle + \langle a^\dagger(0) a^\dagger(\tau) \rangle \langle a(\tau) a(0) \rangle$$

$$+ \langle a^\dagger(0) a(0) \rangle \langle a^\dagger(\tau) a(\tau) \rangle = \langle n \rangle^2 + \langle a^\dagger(0) a(\tau) \rangle \langle a^\dagger(\tau) a(0) \rangle = 0 \text{ (thermal equilibrium)}$$

$$= \langle n \rangle^2 + |\langle a^\dagger(0) a(\tau) \rangle|^2$$